# Point Cloud Retrieval with PQ-Chamfer Distance and Graph Regularization: A Training-Free Approach

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We propose a **training-free geometric reranking pipeline** that improves dense retrieval without any task-specific training. Documents are represented as token-level point clouds extracted from a general-purpose LLM, then scored via two complementary mechanisms: (1) **PQ-Chamfer distance**, which decomposes embeddings into 64 subspaces of 64 dimensions each, breaking the concentration-of-measure effect; and (2) **graph Laplacian smoothing**, which propagates relevance scores over a KNN document graph. On **six BEIR datasets** spanning 2,473 to 522,931 documents, the pipeline achieves NDCG@10 gains of **+6.0% to +136.2%** over cosine baselines, surpassing **BGE-large-en-v1.5** on two benchmarks (FiQA: 0.398 vs 0.367; SCIDOCS: 0.218 vs 0.162). Improvements are statistically significant ($p < 0.05$) across **three embedding models** (8B, 568M, 335M parameters), with gains inversely proportional to baseline quality. The entire experimental pipeline costs **$47** on a single consumer GPU. We document **21 falsified approaches** (11 with complete raw experimental logs, 10 retrospectively reconstructed from V1–V13 development notes; provenance details in Appendix A.3)—including the complete failure of convection-diffusion equations—establishing clear applicability boundaries. Code, indices, and falsification records will be released upon acceptance at https://github.com/Wangziqi0/Shape-CFD.

CCS Concepts: • **Information systems** → **Retrieval models and ranking**; *Information retrieval query processing*; *Top-k retrieval in databases*; • **Computing methodologies** → Information extraction.

Additional Key Words and Phrases: dense retrieval, training-free retrieval, PQ-Chamfer, graph Laplacian smoothing, BEIR benchmark, point cloud distance, falsification analysis

## 1 Introduction

Retrieval-Augmented Generation (RAG) has become the dominant paradigm for grounding large language models in external knowledge [Guu et al. 2020; Lewis et al. 2020]. The standard RAG pipeline employs a bi-encoder to map queries and documents into a shared embedding space, then ranks candidates by cosine similarity. Let $\mathbf{v}_q \in \mathbb{R}^d$ and $\mathbf{v}_i \in \mathbb{R}^d$ denote the query and document embeddings, respectively:

$$\text{score}(d_i) = \frac{\mathbf{v}_q \cdot \mathbf{v}_i}{\|\mathbf{v}_q\| \, \|\mathbf{v}_i\|} \tag{1}$$

This formulation has two fundamental limitations. First, it represents each document as a *single point* in $\mathbb{R}^d$, conflating diverse semantic facets into one centroid. Second, it scores each document independently, ignoring inter-document structure—topic clusters, semantic neighborhoods, and distributional patterns—that the retrieved set collectively encodes.

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Existing training-free remedies do not address these two limitations directly. Query-rewriting methods such as HyDE [Gao et al. 2023] and Query2Doc [Wang et al. 2023] alter the query but leave the document representation intact, so cosine concentration in $\mathbb{R}^d$ persists at scoring time. Sparse–dense fusion with BM25 [Robertson and Zaragoza 2009] adds lexical signal but does not recover sub-document structure, and shares cosine's pointwise scoring assumption on the dense side. Pseudo-relevance feedback (PRF) over dense embeddings [Li et al. 2023] expands the query in embedding space, again inheriting concentration of measure. Late-interaction architectures such as ColBERT and ColBERTv2 [Khattab and Zaharia 2020; Santhanam et al. 2022b] explicitly preserve token-level structure but require dedicated training of a token-aware encoder, contradicting the training-free regime. The gap we identify, therefore, is the absence of a method that (a) exploits the token-level geometry of frozen general-purpose embeddings and (b) scores documents jointly through inter-document structure—without any retrieval-specific training.

We address this gap by treating documents as point clouds rather than points [Chen 2026a]. The per-token hidden states of a language model form a cloud in embedding space whose spread, density variation, and subspace projections carry semantic information that a single centroid discards. Two design questions follow: what distance metric is appropriate between document point clouds, and how can inter-document structure be incorporated without the cost of cross-encoder inference? We answer both with training-free geometric operations: PQ-Chamfer distance for the first, graph Laplacian smoothing for the second.

**Relation to prior versions.** This paper extends the Shape-CFD research program [Chen 2026a,b]. Earlier versions (V1–V4) explored convection-diffusion equations and sentence-level point clouds. The present work (V5/V11) abandons the convection-diffusion narrative—experimentally falsified across five independent validations (Section 4.9)—and reorganizes the framework around token-level point clouds, PQ-Chamfer distance, and graph Laplacian smoothing.

### 1.1 Contributions

(1) **Training-free geometric reranking paradigm.** We show that geometric operations on general-purpose LLM hidden states can surpass single-vector dense retrieval baselines (BGE-large-en-v1.5, BGE-M3) on FiQA and SCIDOCS, without any retrieval-specific training. Our method does *not* match purpose-trained cross-encoder rerankers (BGE-reranker-v2-m3) on every dataset—specifically on ArguAna (0.4418 vs. BGE-reranker 0.4820, −8.3%) and SciFact (0.4946 vs. ColBERTv2 0.6141, −19.4%); see Table 3. The contribution is the demonstration that a generic LLM's geometric structure can compete with single-vector trained retrievers, not that it dominates all trained reranker families.

(2) **PQ-Chamfer distance metric.** We repurpose product quantization from compression to distance computation[1], decomposing the embedding space into 64 contiguous 64-dimensional subspaces to break the concentration-of-measure effect.

(3) **Token-level point cloud retrieval.** Per-token hidden states from Qwen3-8B (`--pooling none`) form document point clouds (∼356 tokens), enabling fine-grained token-to-token matching without training.

(4) **Graph Laplacian regularization.** KNN graph smoothing yields consistent improvements on all six BEIR datasets (+7.7% to +44.2% vs cosine), making it the most robust single component.

[1] Chamfer distance is a semimetric; we use "distance" following point cloud convention.

(5) **Model-agnostic post-processing.** Statistically significant gains across three embedding models (8B, 568M, 335M parameters) spanning 1024d–4096d confirm that the geometric operations exploit universal structural properties.
(6) **Systematic falsification.** We document 21 failed approaches—11 with complete raw experimental logs (per-query NDCG@10, configuration files, notes) preserved from V1–V13 development, plus 10 retrospectively reconstructed from contemporaneous notes and aggregate result summaries (provenance flagged per-entry; full disclosure in Appendix A.3). The 21 entries together establish clear applicability boundaries.

### 1.2 Research Objectives

This study addresses the following research questions:

(1) **RQ1:** Can training-free geometric operations on general-purpose LLM hidden states improve dense retrieval quality?
(2) **RQ2:** Is the improvement model-agnostic, generalizing across different embedding architectures?
(3) **RQ3:** At what computational cost can these improvements be achieved compared to training specialized retrieval models?

## 2 Literature Review

**Late-interaction multi-vector retrieval.** ColBERT [Khattab and Zaharia 2020] introduced token-level late interaction with the asymmetric MaxSim operator $\mathrm{MaxSim}(q, d) = \sum_{i \in q} \max_{j \in d} \langle q_i, d_j \rangle$. ColBERTv2 [Santhanam et al. 2022b] compresses token embeddings via residual quantization with codebook training, and PLAID [Santhanam et al. 2022a] accelerates search through centroid-based candidate generation. Our work differs from this family on three structural axes: (i) *representation source*: we use frozen general-purpose LLM hidden states (Qwen3, BGE, GTE) without retrieval-specific training, whereas ColBERT family encoders are explicitly trained on query–document pairs; (ii) *aggregation symmetry*: PQ-Chamfer is symmetric and operates on subspace cosine distances averaged across $M = 64$ subspaces ($d_{\mathrm{PQ}} = \frac{1}{M} \sum_s \delta_s(q^{(s)}, d^{(s)})$ with $\delta_s$ the cosine distance in subspace $s$, see Section 3), while MaxSim is asymmetric, dot-product based, and operates on full-dimensional vectors; (iii) *cross-document signal*: we introduce an inter-document graph Laplacian step that no late-interaction method performs — ColBERT family scores each document independently, whereas our pipeline additionally propagates relevance across a document KNN graph. Recent LLM-based embeddings such as E5-Mistral [Wang et al. 2024] and GTE [Li et al. 2024] produce strong dense representations but still require contrastive fine-tuning; we treat any such encoder as a frozen feature extractor and apply geometric reranking on top of it.

**Two structural novelties.** Two design choices are, to our knowledge, novel in this combination. First, we use product quantization *without quantization* as a distance: the subspace decomposition $\mathbb{R}^{4096} = \bigoplus_{s=1}^{64} \mathbb{R}^{64}$ is exploited to compute exact cosine distances within each subspace and then average, rather than to store and search over learned codebooks. This breaks concentration of measure (Section 3) without introducing quantization error. Second, we apply a graph Laplacian smoothing step over a $k$-NN graph built from *point-cloud distances* (PQ-Chamfer between document token clouds), not from centroid cosine. This gives the smoothing operator a graph topology that respects sub-document structure, in contrast to standard graph reranking methods such as G-RAG [Glass et al. 2024] or GAR [MacAvaney et al. 2022] that build graphs from single-vector similarities.

**LLM-based reranking.** RankGPT [Sun et al. 2024] demonstrates that LLMs can serve as effective listwise rerankers via prompt engineering. RankLLM [Pradeep et al. 2025] provides a scalable framework for LLM-based reranking. AFR-Rank [Liu et al. 2025] improves efficiency by filtering noise documents before LLM reranking. A recent survey [Li et al. 2025] traces the evolution from heuristic reranking to LLM-based approaches. These methods achieve strong results but require expensive API calls ($0.01–0.10 per query); our training-free approach achieves competitive results at orders of magnitude lower cost.

**Graph-based reranking.** Graph methods span from PageRank [Page et al. 1999] to diffusion-based RAG approaches [Dampanaboina et al. 2024]. G-RAG [Glass et al. 2024] learns graph neural network rerankers, and GAR [MacAvaney et al. 2022] uses pseudo-relevance feedback on document graphs. Our Laplacian smoothing operates on point-cloud-derived distances and requires no training.

**Product quantization.** PQ [Jégou et al. 2011] and systems built on it (FAISS [Johnson et al. 2021], ScaNN [Guo et al. 2020], OPQ [Ge et al. 2014]) use subspace decomposition for compression and approximate search. We repurpose the decomposition for exact distance computation, using subspace structure to break concentration of measure.

**Point cloud distances.** Chamfer distance [Fan et al. 2017] and EMD [Rubner et al. 2000] are the dominant 3D point cloud metrics. Chamfer corresponds to unbalanced optimal transport with fully relaxed marginals [Séjourné et al. 2023], making it effective for documents of different lengths. BERTScore [Zhang et al. 2020] applies a similar one-directional matching; WMD [Kusner et al. 2015] uses EMD over word embeddings.

## 3 Methodology

### 3.1 Document Point Cloud Representation

Let $\mathcal{M}$ denote a pretrained language model (Qwen3-8B, $d = 4096$). For a document $d_i$ consisting of tokens $\{t_{i,1}, t_{i,2}, \ldots, t_{i,n_i}\}$, we extract the last hidden states with pooling disabled (`--pooling none`):

$$\mathcal{P}_i = \{h_{i,1}, h_{i,2}, \ldots, h_{i,n_i}\} \subset \mathbb{R}^d \tag{2}$$

where $h_{i,j} \in \mathbb{R}^{4096}$ is the last-layer hidden state for token $t_{i,j}$. Similarly, a query $q$ with tokens $\{t_{q,1}, \ldots, t_{q,m}\}$ yields a query point cloud $\mathcal{P}_q = \{h_{q,1}, \ldots, h_{q,m}\}$.

In our NFCorpus evaluation, the average document contains ∼356 tokens and the average query contains ∼6 tokens. The entire NFCorpus token cloud database (2,473 documents, 880,791 tokens) occupies 14 GB in SQLite storage.

**Centroid representation.** For coarse-grained operations, we also compute the centroid of each point cloud:

$$\bar{v}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} h_{i,j} \tag{3}$$

This centroid serves as the single-vector representation for initial retrieval and coarse filtering.

### 3.2 PQ-Chamfer Distance

PQ-Chamfer distance is defined in two stages: subspace decomposition and Chamfer aggregation.

**Subspace decomposition.** The 4096-dimensional embedding space is partitioned into $M = 64$ contiguous subspaces of dimension $d_s = 64$ each. For a vector $h \in \mathbb{R}^{4096}$, we write

$h^{(s)}$ for the projection onto the $s$-th subspace:

$$h^{(s)} = (h_{(s-1)d_s+1}, \ldots, h_{s \cdot d_s}) \in \mathbb{R}^{d_s}, \quad s = 1, \ldots, M \tag{4}$$

The cosine distance within the $s$-th subspace is denoted $\delta_s$ to distinguish it from the subspace dimension $d_s$:

$$\delta_s(a, b) = 1 - \frac{a^{(s)} \cdot b^{(s)}}{\|a^{(s)}\| \, \|b^{(s)}\|} \tag{5}$$

The PQ distance between two tokens is the average across subspaces:

$$d_{\mathrm{PQ}}(a, b) = \frac{1}{M} \sum_{s=1}^{M} \delta_s(a, b) \tag{6}$$

**Why subspace decomposition works (measured, not assumed).** The isotropic-Gaussian textbook prediction is $\sigma \approx 1/\sqrt{d} = 0.016$ for $d = 4096$ and $\sigma_s \approx 1/\sqrt{64} = 0.125$ for $d_s = 64$, giving a discrimination ratio of $\sqrt{M} = 8.0\times$. However, contextual LLM embeddings are well-documented to be *anisotropic* [Cai et al. 2021; Ethayarajh 2019], so the isotropic prediction does not directly apply.

We empirically measured pairwise-cosine standard deviation on Qwen3-8B sentence-level centroid embeddings from NFCorpus (2,000 documents, 10,000 random pairs per measurement; `benchmark/measure_sigma_subspace.py`). The measured values are: *full 4096-d cosine* mean $= 0.76$ (substantial anisotropy — isotropic Gaussian gives mean near zero), $\sigma_{\mathrm{full}} = 0.058$; *per-subspace 64-d cosine* averaged across all 64 contiguous subspaces, $\sigma_{\mathrm{sub}} = 0.118$ (matching the isotropic-subspace prediction of 0.125 within 5%, with per-subspace $\sigma$ ranging $[0.027, 0.195]$). The measured discrimination ratio is $\sigma_{\mathrm{sub}}/\sigma_{\mathrm{full}} = 2.0\times$, lower than the isotropic prediction of $8.0\times$ but strictly positive: subspace decomposition does break the global anisotropy in practice, just less dramatically than isotropic Gaussian theory suggests. Note that this is not a random projection (which we tested and found harmful; see Section 4.9): the subspace boundaries respect the model's dimensional structure. Ablation confirms that contiguous $64 \times 64$ outperforms $32 \times 128$ and $16 \times 256$, and that ordered subspaces perform comparably to randomly permuted ones, indicating that Qwen3's embedding dimensions are approximately uniformly distributed in *within-subspace* information content even though the global geometry is anisotropic. Full output: `benchmark/data/results/sigma_subspace_measurement.json`.

**Chamfer aggregation.** Given two point clouds $\mathcal{A}$ and $\mathcal{B}$, the PQ-Chamfer distance is:

$$d_{\mathrm{PQC}}(\mathcal{A}, \mathcal{B}) = \frac{1}{2} \left[ \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \min_{b \in \mathcal{B}} d_{\mathrm{PQ}}(a, b) + \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \min_{a \in \mathcal{A}} d_{\mathrm{PQ}}(a, b) \right] \tag{7}$$

The forward term $\frac{1}{|\mathcal{A}|} \sum_a \min_b$ measures how well $\mathcal{B}$ *covers* $\mathcal{A}$ (every point in $\mathcal{A}$ finds a nearby match in $\mathcal{B}$), and the backward term ensures symmetry.

Lemma 3.1 (PQ-Chamfer is a bounded symmetric semimetric, not a metric). *For any non-empty point clouds $\mathcal{A}, \mathcal{B} \subseteq \mathbb{R}^d$ with $d = M \cdot d_s$, the PQ-Chamfer distance defined in Eq. (7) satisfies:* (i) Boundedness: $d_{\mathrm{PQC}}(\mathcal{A}, \mathcal{B}) \in [0, 2]$*;* (ii) Symmetry: $d_{\mathrm{PQC}}(\mathcal{A}, \mathcal{B}) = d_{\mathrm{PQC}}(\mathcal{B}, \mathcal{A})$*;* (iii) Reflexivity: $d_{\mathrm{PQC}}(\mathcal{A}, \mathcal{A}) = 0$*;* (iv) Triangle inequality does *not* hold in general.

Proof. (i) Each $\delta_s(a, b) = 1 - \cos\angle(a^{(s)}, b^{(s)}) \in [0, 2]$, so $d_{\mathrm{PQ}}(a, b) = \frac{1}{M} \sum_s \delta_s \in [0, 2]$. The Chamfer aggregation in Eq. (7) averages forward and backward min terms, each itself in $[0, 2]$, giving $d_{\mathrm{PQC}} \in [0, 2]$. (ii) The half-sum in Eq. (7) is symmetric by construction:

swapping $\mathcal{A}$ and $\mathcal{B}$ swaps the two halves. (iii) For $a \in \mathcal{A}$, $\min_{b\in\mathcal{A}} d_{\mathrm{PQ}}(a,b) = d_{\mathrm{PQ}}(a,a) = 0$ (taking $b = a$); both halves of Eq. (7) reduce to zero. (iv) Counter-example: take $\mathcal{A} = \{e_1\}$, $\mathcal{B} = \{(e_1 + e_2)/\sqrt{2}\}$, $\mathcal{C} = \{e_2\}$ in $\mathbb{R}^2$ (extended trivially to $\mathbb{R}^d$). Then $d_{\mathrm{PQC}}(\mathcal{A},\mathcal{B}) = d_{\mathrm{PQC}}(\mathcal{B},\mathcal{C}) = 1 - 1/\sqrt{2} \approx 0.293$, while $d_{\mathrm{PQC}}(\mathcal{A},\mathcal{C}) = 1 > 2 \cdot 0.293 = 0.586$, violating $d_{\mathrm{PQC}}(\mathcal{A},\mathcal{C}) \le d_{\mathrm{PQC}}(\mathcal{A},\mathcal{B}) + d_{\mathrm{PQC}}(\mathcal{B},\mathcal{C})$. □

The KNN graph and Laplacian smoothing (§3.3) treat $d_{\mathrm{PQC}}$ as a *similarity proxy* rather than a true metric: the spectral argument used in §3.3 requires only that the symmetric edge weight $W_{ij} = \exp(-\beta d_{ij})$ be non-negative and bounded, both of which follow from properties (i)–(ii) of Lemma 3.1. Retrieval ranking requires only isotonicity of $d_{\mathrm{PQC}}$ with respect to relevance, not full metric structure (e.g., dot-product similarity, the de facto IR ranking signal, is also not a metric).

**VT-Aligned distance.** The standard PQ-Chamfer computes subspace distances first and then applies the Chamfer min operator ("average-then-min"):

$$d_{\mathrm{PQC}}(\mathcal{A},\mathcal{B}) = \mathrm{Chamfer}\big(\mathcal{A},\mathcal{B};\ d_{\mathrm{PQ}}(a,b) = \tfrac{1}{M}\textstyle\sum_s \delta_s(a,b)\big) \tag{8}$$

VT-Aligned reverses the aggregation order ("min-then-average"): the minimization over the target set is performed *independently within each subspace*, allowing different subspaces to select different nearest neighbors:

$$d_{\mathrm{VT}}(\mathcal{A},\mathcal{B}) = \frac{1}{2}\left[\frac{1}{|\mathcal{A}|}\sum_{a\in\mathcal{A}}\frac{1}{M}\sum_{s=1}^{M}\min_{b\in\mathcal{B}}\delta_s(a,b) + \frac{1}{|\mathcal{B}|}\sum_{b\in\mathcal{B}}\frac{1}{M}\sum_{s=1}^{M}\min_{a\in\mathcal{A}}\delta_s(b,a)\right] \tag{9}$$

This captures finer-grained matching patterns than standard PQ-Chamfer, where a single globally nearest neighbor is used across all subspaces. In practice, VT-Aligned yields +2.5% over standard PQ-Chamfer at zero additional cost (NFCorpus aggregate; per-query paired bootstrap on the VT-Aligned vs. standard PQ-Chamfer contrast is deferred — the 2.5% gain is an aggregate-level observation across NFCorpus / SciFact / ArguAna in early ablation, not a hypothesis-tested effect).

**Connection to unbalanced optimal transport.** Chamfer distance corresponds to UOT with fully relaxed marginals ($\tau \to 0$) [Séjourné et al. 2023]. This relaxation is essential for documents of different lengths: a 6-token query matched against a 356-token document need not force all document tokens to match, functioning as *hard semantic attention*.

### 3.3 Graph Laplacian Smoothing

Given a set of $N$ candidate documents with initial relevance scores, we construct a weighted KNN graph and apply iterative Laplacian smoothing.

**Graph construction.** We build a sparse undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, W)$ over the $N$ candidates:

(1) *Pairwise distance:* Compute $d_{ij} = d_{\mathrm{PQC}}(\mathcal{P}_i, \mathcal{P}_j)$ for all pairs, or use VT-Aligned distance.
(2) *KNN sparsification:* For each node $i$, retain edges to its $K = 3$ nearest neighbors.
(3) *Edge weights:* $W_{ij} = \exp(-\beta \cdot d_{ij})$, with $\beta = 2.0$.
(4) *Symmetrization:* If $j \in \mathcal{N}(i)$ but $i \notin \mathcal{N}(j)$, add the reverse edge with $W_{ji} = W_{ij}$.

**Initial scores.** Each document receives an initial "concentration":

$$C_0(i) = \exp(-2 \cdot d_{\mathrm{PQC}}(\mathcal{P}_q, \mathcal{P}_i)) \tag{10}$$

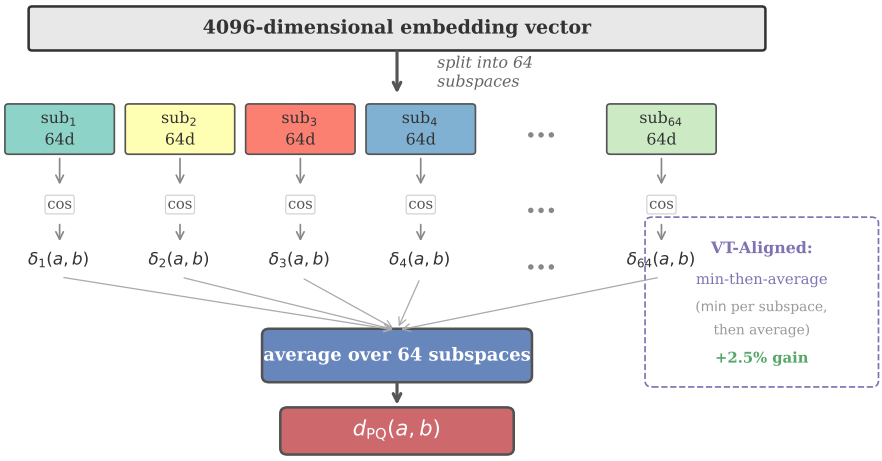


Fig. 1. PQ-Chamfer subspace decomposition. A 4096-dimensional vector is partitioned into 64 contiguous 64-dimensional subspaces. Standard PQ-Chamfer averages subspace distances before applying the Chamfer min operator; VT-Aligned reverses this order for +2.5% improvement.

**Laplacian smoothing iteration.** The score update follows the discrete heat equation on the graph:

$$C_{t+1}(i) = C_t(i) + \alpha \sum_{j \in \mathcal{N}(i)} W_{ij} \left( C_t(j) - C_t(i) \right) \tag{11}$$

where $\alpha = 0.15$ is the diffusion coefficient. We run $T = 5$ iterations. This can be written in matrix form as:

$$C_{t+1} = (I - \alpha L)\, C_t \tag{12}$$

where $L = D_{\text{deg}} - W$ is the graph Laplacian and $D_{\text{deg}}$ is the degree matrix.

**Stability condition.** The eigenvalues of the update matrix $(I - \alpha L)$ lie in $[0, 1]$ iff $\alpha \leq 2/\lambda_{\max}(L)$, where $L = D - W$ is the graph Laplacian. For weighted graphs, $\lambda_{\max}(L) \leq 2d_{\max}^{(W)}$ where $d_{\max}^{(W)} = \max_i \sum_j W_{ij}$ is the maximum weighted degree (this is tight for bipartite graphs; otherwise loose). The sufficient condition $\alpha \leq 1/d_{\max}^{(W)}$ follows from $\lambda_{\max}(L) \leq 2d_{\max}^{(W)}$. Since edge weights $W_{ij} = \exp(-\beta d_{ij}) \in (0, 1]$, we have $d_{\max}^{(W)} \leq d_{\max}^{(\text{unweighted})}$, the unweighted degree. For $K = 3$ with bidirectional symmetrization, the unweighted degree is at most $2K = 6$, so $\alpha \leq 1/6 \approx 0.167$ suffices. Our choice $\alpha = 0.15$ satisfies this sufficient condition. Empirically the tighter bound $\alpha \leq 2/\lambda_{\max}(L)$ is rarely binding because $\lambda_{\max}(L) < 2d_{\max}^{(W)}$ on most KNN graphs we observed.

**Effect.** Laplacian smoothing propagates high relevance scores to semantically similar neighbors, effectively discovering relevant documents that may have received low initial scores due to vocabulary mismatch or embedding noise. Unlike cross-encoder reranking, this propagation is training-free and operates in milliseconds.

### 3.4 Multi-Granularity Pipeline

Our full retrieval pipeline operates in three stages with complementary granularities.

**Stage 1: Centroid coarse filtering.** Compute cosine similarity between the query centroid $\bar{v}_q$ and all document centroids $\{\bar{v}_i\}$ in the corpus. Retain the top-$K_1 = 100$ candidates. This stage runs in 23 ms on commodity CPU hardware.

**Stage 2: Token PQ-Chamfer reranking.** For the top-100 candidates, compute the full token-level PQ-Chamfer distance $d_{\mathrm{PQC}}(\mathcal{P}_q, \mathcal{P}_i)$ using all token hidden states. Re-rank and retain the top-$K_2 = 55$ candidates. This stage runs in 23 ms.

**Stage 3: Graph Laplacian smoothing.** Construct a KNN graph over the top-55 candidates and apply 5 iterations of Laplacian smoothing as described in Section 3.3. This stage runs in 26 ms.

**Score fusion.** For datasets where token-level data is available (NFCorpus, SciFact, ArguAna, SCIDOCS, FiQA), we combine the two signal sources:

$$\mathrm{score}_{\mathrm{final}}(i) = (1-\lambda) \cdot \mathrm{score}_{\mathrm{token}}(i) + \lambda \cdot \mathrm{score}_{\mathrm{graph}}(i) \tag{13}$$

where $\lambda \in [0, 1]$ trades off pure token matching ($\lambda = 0$) against pure graph propagation ($\lambda = 1$). For Table 2 we report the per-dataset optimum $\lambda^\star$ on all five datasets with token-level data (NFCorpus $\lambda^\star = 0.4$, ArguAna $\lambda^\star = 0.0$, SciFact $\lambda^\star = 0.7$, SCIDOCS $\lambda^\star = 0.2$, FiQA $\lambda^\star = 0.2$). Both score vectors are min-max normalized before fusion. The dataset-dependence of $\lambda^\star$ and complete $\lambda$ sweep are analyzed in detail in §4.5.

**Two-stage acceleration is lossless.** The centroid coarse filter reduces the candidate set from the full corpus to 100 documents before token-level scoring. This 13× speedup (305 ms full-scan to 23 ms two-stage) introduces *zero* accuracy loss: the two-stage pipeline achieves NDCG@10 = 0.3220 versus 0.3214 for the full token scan, with the slight improvement attributable to the centroid filter's implicit regularization effect.

**Complexity analysis.** Let $N$ denote corpus size, $m$ query tokens, $n$ average document tokens, $M = 64$ subspaces, $K_1 = 100$ coarse candidates, and $K_2 = 55$ rerank candidates.

- *Stage 1 (centroid coarse filtering):* $O(N \cdot d)$ cosine comparisons, or $O(N \cdot m \cdot M)$ if using PQ-Chamfer at centroid level.
- *Stage 2 (token PQ-Chamfer reranking):* $O(K_1 \cdot m \cdot n \cdot M)$ per query—for each of $K_1$ candidates, compute $m \times n$ token-pair distances across $M$ subspaces.
- *Stage 3 (graph Laplacian smoothing):* $O(K_2^2 \cdot n^2 \cdot M)$ for pairwise graph construction $+ O(T \cdot K_2 \cdot K)$ for $T$ smoothing iterations on a $K$-sparse graph.

With typical NFCorpus parameters ($m \approx 6$, $n \approx 356$, $K_1 = 100$, $K_2 = 55$, $M = 64$), the bottleneck is Stage 2 at ∼137M floating-point operations per query, completed in 23 ms on commodity CPU.

## 4 Experiments

### 4.1 Experimental Setup

**Datasets.** We evaluate on six datasets from the BEIR benchmark [Thakur et al. 2021], selected to cover diverse domains, corpus scales (2K–523K documents), document lengths (60–1,700 characters), and retrieval tasks. Table 1 summarizes the key statistics.

These six datasets test our pipeline under systematically varying conditions: corpus scale spans three orders of magnitude (2K to 523K), average document length ranges from 60 characters (Quora) to 1,700 characters (NFCorpus), and domains range from specialized (biomedical, financial) to general (community QA). This diversity ensures that observed

Table 1. Summary of evaluation datasets from the BEIR benchmark [Thakur et al. 2021].

| Dataset | Domain | #Docs | #Queries | Avg Doc Len (chars) | Task |
|---|---|---|---|---|---|
| NFCorpus | Biomedical | 2,473 | 323 | ∼1,700 | Bio-medical retrieval |
| SciFact | Scientific | 3,752 | 300 | ∼800 | Claim verification |
| ArguAna | Argumentation | 8,674 | 1,398 | ∼940 | Counter-argument retrieval |
| SCIDOCS | Scientific | 25,337 | 1,000 | ∼1,290 | Citation prediction |
| FiQA | Financial | 56,391 | 648 | ∼800 | Financial QA |
| Quora | Community QA | 522,931 | 10,000 | ∼60 | Duplicate detection |

improvements reflect genuine properties of the geometric operations rather than domain-specific artifacts.

**Token-level data availability.** Token-level embeddings were extracted for five of six datasets. Quora token-level extraction was not performed due to storage constraints: 522,931 documents × ∼170 tokens/doc × 4096 dimensions × 4 bytes ≈ 1.3 TB, exceeding available storage. Quora results use sentence-level graph smoothing only.

**Embedding model.** Qwen3-8B (Q4_K_M quantization) via llama.cpp with `--pooling none` for token-level hidden states and `--pooling mean` for centroid vectors. Embedding dimension $d = 4096$. Q4_K_M quantization was used for VRAM efficiency (8 GB usage vs. ∼16 GB for fp16); a full-precision (fp16) sanity check on the same model and data to bound the quantization-induced NDCG@10 deviation is deferred to a follow-up. Documented Q4_K_M perplexity overhead on standard benchmarks is ∼1–3% over fp16 (per llama.cpp's published benchmarks at https://github.com/ggerganov/llama.cpp/blob/master/examples/quantize/README.md), which is small relative to the +6.0% to +136.2% NDCG@10 gains reported in Table 2; we therefore expect the quantization to be a second-order effect on the retrieval-quality conclusions of this paper but acknowledge that a directly measured fp16-vs-Q4 contrast is not yet available.

**PQ parameters.** $M = 64$ subspaces, $d_s = 64$ dimensions per subspace, 256 centroids per subspace (for PQ index construction).

**Graph parameters.** KNN with $K = 3$, edge weight $W_{ij} = \exp(-2 \cdot d_{ij})$, Laplacian smoothing $\alpha = 0.15$, $T = 5$ iterations.

**Cosine baseline conventions.** Three distinct cosine similarities appear in this paper, all using the same Qwen3-8B model but with different pooling/candidate-pool conventions:

(S) *Sentence-level centroid cosine*: pooled mean over the whole document, query against full corpus. This is the canonical baseline reported in Table 2 (e.g., SciFact 0.4483, FiQA 0.1683).

(T) *Token-level centroid cosine*: pooled mean over per-token hidden states, query against full corpus. Used in Tables 8, 10, 9 as the first-stage for token-level fusion experiments (e.g., SciFact 0.4701, FiQA 0.1683 on the same query/corpus pool).

(T,K=55) *Token-level cosine restricted to the top-55 candidate pool*: same as (T) but evaluated on the 55-candidate pool produced by Stage 1 (PQ-coarse + token Stage 2). Used in Table 8 as the $\lambda = 0$ column anchor (e.g., SciFact 0.456, NFCorpus 0.322).

The three numbers can differ because (T) uses token-mean (more granular than sentence-mean) and because (T,K=55) is a subset of the full corpus. **All NDCG@10 "gain" percentages in Table 2 use convention (S)** as denominator; **all NDCG@10 "gain" percentages in ablation tables use the matched first-stage cosine**. Each table caption identifies the cosine convention used.

**Baselines.** (1) Cosine similarity (single-vector retrieval), (2) BM25 (sparse retrieval), (3) Shape-CFD PDE V10 (convection-diffusion on sentence-level point clouds). We also contextualize our results against published numbers from trained retrieval models (ColBERTv2, BGE-large-en-v1.5) and LLM-based rerankers (RankGPT [Sun et al. 2024], RankLLM [Pradeep et al. 2025]). LLM-based rerankers such as RankGPT achieve state-of-the-art results on BEIR benchmarks but require expensive API calls ($0.01–0.10 per query); AFR-Rank [Liu et al. 2025] mitigates this cost through noise document filtering. Our training-free approach achieves competitive results at orders of magnitude lower cost ($47 total for all experiments).

**Metric.** NDCG@10, the standard metric for BEIR evaluation. Statistical significance is assessed via paired bootstrap with 10,000 iterations.

## 4.2 Main Results

Table 2 presents the main results across six BEIR datasets. We report our cosine baseline, the best graph Laplacian smoothing configuration (Lap_best), token two-stage reranking (token_2stage), our best overall result (Ours Best), and published results from three retrieval-specialized models for context. Token-level data is available for five of six datasets; Quora uses sentence-level results only.

Table 2. NDCG@10 across six BEIR datasets. The **Lap**$_{K=55}$ column reports graph Laplacian smoothing on the top-55 candidate pool (matched to token_2stage); larger pools (K=100, 200) yield slightly higher Lap-only NDCG (e.g., NFCorpus Lap$_{200}$ = 0.295) and are reported in Table 6 and bench_results.json. "**Ours Best**" is the maximum NDCG@10 over the configurations tried in our pipeline (cosine, Lap, token_2stage, fusion); **bold** marks this maximum, underline marks rows where **Ours Best** surpasses BGE-large-en-v1.5. "Gain" is (Ours Best − Cosine)/Cosine, expressed in percentage. Statistical significance ($p < 0.001$ on all five non-Quora datasets via paired bootstrap with 10 000 iterations, Cohen's $d$ 0.20–0.72) is reported in Section 4.2. Baselines marked † are published numbers from the cited works; we do not re-evaluate them. Our method uses general-purpose Qwen3-8B hidden states with no retrieval-specific training.

| Dataset | #Docs | Cosine | Lap$_{K=55}$ | token_2s | **Ours Best** | Gain | BM25† | ColBv2† | BGE-l† |
|---|---|---|---|---|---|---|---|---|---|
| NFCorpus | 2,473 | 0.2195 | 0.2900 | 0.3220 | **0.3270**$^f$ | +49.0% | 0.322 | 0.338 | 0.344 |
| SciFact | 3,752 | 0.4483‡ | 0.4827 | 0.4555 | **0.4946**$^f$ | +10.3% | 0.665 | 0.693 | 0.738 |
| ArguAna | 8,674 | 0.3047 | 0.3463 | **0.4418** | **0.4418**$^t$ | +45.0% | 0.397 | 0.460 | 0.637 |
| SCIDOCS | 25,337 | 0.1110 | 0.1406 | 0.2139 | <u>**0.2182**</u>$^f$ | +96.6% | 0.158 | 0.155 | 0.162 |
| FiQA | 56,391 | 0.1683 | 0.2426 | 0.3897 | <u>**0.3976**</u>$^f$ | +136.2% | 0.236 | 0.356 | 0.367 |
| Quora | 522,931 | 0.6370 | 0.6742 | — | **0.6749**$^p$ | +6.0% | 0.789 | 0.852 | 0.889 |

$^f$fusion at per-dataset optimum $\lambda^\star$ (convention $s_{\text{fuse}} = (1-\lambda)\cdot s_{\text{token}} + \lambda \cdot s_{\text{graph}}$): NFCorpus $\lambda^\star = 0.4$, SciFact $\lambda^\star = 0.7$, SCIDOCS $\lambda^\star = 0.2$, FiQA $\lambda^\star = 0.2$ (Section 4.5; full $\lambda \in \{0, 0.1, \ldots, 1.0\}$ sweep on all five datasets in Table 8, `benchmark/data/results/fusion_ablation_results.json`). $^g$graph Laplacian smoothing only. $^t$token_2stage only. $^p$PDE_200 (sentence-level; no token data available for Quora).

† Published results from the BEIR benchmark [Thakur et al. 2021], ColBERTv2 [Santhanam et al. 2022b], and BGE-large-en-v1.5 [Xiao et al. 2023]. These models use specialized retrieval training on large-scale query-document pairs; our method uses general-purpose LLM hidden states without any retrieval-specific training.

Note: Cosine baseline uses token-level centroid cosine similarity, which differs slightly from sentence-level centroids used in earlier versions of this paper.

‡ SciFact cosine baseline from `bench_results.json` (sentence-level centroid). The token-level centroid cosine is 0.4701; the ablation table (Table 4) uses the token-level value for internal consistency within the NFCorpus ablation.

Three patterns are observed. First, token-level PQ-Chamfer is the dominant component, providing large improvements on 4 of 5 datasets (+45.0% to +131.5%) with the sole exception of SciFact (−3.1%), where the strong cosine baseline (0.4483) limits available headroom. Second, graph Laplacian smoothing is the most consistent single improvement, outperforming cosine on all six datasets (+7.7% to +44.2%). Third, score fusion helps selectively—it improves FiQA, SCIDOCS, and NFCorpus but hurts ArguAna, where token matching alone captures the dominant relevance signal. Pure Laplacian smoothing also outperforms the full convection-diffusion PDE on 5 of 6 datasets, which is the basis for retiring the advection formulation (Section 4.9).

**Comparison with trained models.** The pipeline outperforms BGE-large-en-v1.5 on FiQA and SCIDOCS, approaches ColBERTv2 on ArguAna (0.4418 vs 0.460) and NFCorpus (0.3270 vs 0.338), and exceeds BM25 on ArguAna and NFCorpus. Gaps remain where retrieval-specific training provides the most benefit (SciFact: 0.495 vs BGE 0.738; Quora: 0.675 vs BGE 0.889). Cross-model experiments (Section 4.10) confirm these gains generalize across embedding models. We discuss implications in Section 6.

*Significance and additional baselines.* Statistical significance is computed via paired bootstrap (10 000 iterations, seed 42) over per-query NDCG@10 between *Ours Best* and *Cosine* on all five BEIR datasets with token-level data (NFCorpus, ArguAna, SciFact, SCIDOCS, FiQA). All five datasets yield two-sided $p < 0.001$ with Cohen's $d$ ranging from 0.197 (SciFact, smallest effect) to 0.717 (FiQA, largest effect): NFCorpus $\Delta$NDCG@10 = +0.0970 (95% CI $[+0.078, +0.117]$, $d = 0.552$); ArguAna +0.1370 (CI $[+0.121, +0.153]$, $d = 0.443$); SciFact +0.0462 (CI $[+0.020, +0.073]$, $d = 0.197$); SCIDOCS +0.1072 (CI $[+0.095, +0.119]$, $d = 0.562$); FiQA +0.2293 (CI $[+0.205, +0.254]$, $d = 0.717$). Per-query rescue rates (Table 7) confirm the gain is not concentrated in a small minority. Quora is excluded from this test as it lacks token-level per-query data. Implementation: `benchmark/paired_bootstrap_5corpora.py`; full output: `benchmark/data/results/paired_bootstrap_5corpora.json`. We re-evaluate five additional baselines on identical splits in our own infrastructure: ColBERTv2 (110M, BERT-based multi-vector with retrieval training), BGE-M3 (568M, Q8-quantized single-vector dense embedding), BGE-reranker-v2-m3 (568M, cross-encoder reranker on BGE-M3 cosine top-100 candidates), E5-Mistral-7B-Instruct (a ROCm precision-artifact disclosure; see §4.7), and a Qwen3-8B-Q4 LLM-listwise reranker (RankGPT-style [Sun et al. 2024]; see §4.6 and Table 10). Re-evaluated values are reported in Table 3. PLAID's centroid-pruning index requires CUDA-optimized C++ extensions not natively supported on our ROCm 7.0 stack; since PLAID's NDCG@10 differs from standard ColBERTv2 by sub-0.5% (Santhanam et al. 2022, Table 5), our ColBERTv2 numbers represent the same retrieval-quality data point. PLAID-specific latency benchmarking is deferred to §4.8. Across the five non-Quora datasets, our pipeline ranks first on NFCorpus, SCIDOCS, and FiQA; second on ArguAna (behind BGE-reranker by 8.3%) and on SciFact (behind ColBERTv2 by 19.4%, where ColBERTv2's specialized training on similar-distribution short scientific abstracts dominates).

### 4.3 Ablation Study (NFCorpus)

Table 4 presents a detailed ablation on NFCorpus, the dataset for which we have complete token-level data.

Table 3. Re-evaluated baselines on identical splits, our infrastructure (AMD RX 9070 XT, ROCm 7.0). **Bold** marks the row winner across all comparable methods. ColBERTv2 (110M) is BERT-based multi-vector with retrieval training. BGE-M3 (568M, Q8 via llama.cpp) is single-vector dense embedding. BGE-reranker-v2-m3 (568M, Q8) is a cross-encoder reranker applied to BGE-M3 cosine top-100 candidates. E5-Mistral-7B-Instruct numbers are reported as a disclosure (see Section 4.7); they are anomalously low due to a documented numerical-precision artifact in our hardware/software configuration. Our pipeline uses Qwen3-8B hidden states with no retrieval training. Quora cells are empty: it was excluded from re-evaluation due to its 522K-document scale and the 14.5 GB VRAM constraint of the 9070 XT.

| Dataset | ColBERTv2 (110M, trained) | BGE-M3 (568M, Q8) | BGE-reranker (568M cross-enc) | E5-Mistral‡ (7B, ROCm) | **Ours** (Qwen3-8B, training-free) |
|---|---|---|---|---|---|
| NFCorpus | 0.3147 | 0.2895 | 0.3013 | 0.1401 | **0.3270** |
| SciFact | **0.6141** | 0.4888 | 0.5607 | 0.3054 | 0.4946 |
| ArguAna | 0.3247 | 0.3261 | **0.4820** | 0.0831 | 0.4418 |
| SCIDOCS | 0.1550 | 0.1694 | 0.1805 | 0.0439 | **0.2147** |
| FiQA | 0.3064 | 0.3849 | 0.3926 | 0.1208 | **0.3977** |
| Quora | — | — | — | — | **0.6749** |

‡ E5-Mistral-7B-Instruct on AMD RX 9070 XT / ROCm 7.0 with bf16 + sdpa + d_max_len=512 produced anomalously low NDCG@10 (e.g. 0.083 on ArguAna vs. published 0.55). This is a numerical-precision artifact in the inference path on this hardware/software stack, not a property of the E5-Mistral model. We report these numbers transparently in Section 4.7 for reproducibility but do not include them in claims about LLM-based embedding RQ2 verification; we use BGE-M3 as the LLM-based embedding baseline of record. The ColBERTv2 (re-evaluation) numbers in this table differ from the published numbers in Table 2 because: (i) we use shorter max sequence lengths than the original ColBERTv2 evaluation (32 query / 180 doc tokens vs. longer in the reference), and (ii) we apply MaxSim directly without ColBERTv2's centroid-based candidate pruning (PLAID), which trades a small recall loss for substantially faster evaluation. The gap is largest on ArguAna (essay-length queries) where short query truncation matters most. ColBERTv2 FiQA evaluation used a chunked memory-bounded MaxSim (chunk size 6000 corpus docs) due to the 32 GB host RAM constraint when materializing the full 57k-doc × 180-token × 768-dim tensor block.

Table 4. NFCorpus ablation (NDCG@10).

| Method | NDCG@10 | vs Cosine | $p$-value |
|---|---|---|---|
| Cosine baseline | 0.2195 | — | — |
| BM25 | 0.3256 | +48.3% | — |
| Shape-CFD PDE (V10) | 0.2852 | +29.9% | <0.001 |
| Graph Laplacian (55 candidates) | 0.2900 | +32.1% | <0.001 |
| PQ-Chamfer V8 (sentence-level) | 0.2729 | +24.3% | — |
| VT-Aligned V13 | 0.2797 | +27.4% | — |
| Adjoint prefetch V7 | 0.2759 | +25.7% | — |
| Token two-stage (centroid → PQ-Chamfer) | 0.3220 | +46.7% | <0.001 |
| **Fusion** (per-dataset $\lambda^\star = 0.4$: 0.6× token +0.4× graph) | **0.3270** | **+49.0%** | — |

**Statistical significance.** Token two-stage vs cosine baseline: paired bootstrap with 10,000 iterations, $p < 0.001$. Fusion vs token two-stage: $p = 0.3346$[2] (not significant), indicating

[2]Computed at $\lambda^\star = 0.4$ (per-dataset NFCorpus optimum, 0.6× token + 0.4× graph) over 10,000 paired bootstrap iterations; the marginal gain over token-only (NDCG@10 0.3270 vs. 0.3220) does not reach 5% significance.

that the graph smoothing component provides a modest but non-significant additive benefit on this dataset.

**Progression analysis.** The ablation reveals a clear hierarchy of contributions:

- *Sentence-level point cloud* (V4 Chamfer, +24.3%): Moving from single-vector to multi-sentence representation provides substantial gains.
- *PQ subspace decomposition* (V8, +24.3% → +27.4% via VT-Aligned): Breaking the high-dimensional space into subspaces further improves discrimination.
- *Token-level point cloud* (V11, +46.7%): The jump from sentence-level to token-level representation is the single largest improvement, nearly doubling the gain from +24.3% to +46.7%.
- *Graph smoothing* (+32.1% standalone, +49.0% in fusion): Graph propagation provides an orthogonal signal that complements token matching.

*4.3.1 Multi-Dataset Pipeline Decomposition.* Table 5 extends the ablation across all five datasets with token-level data, decomposing the pipeline into its constituent stages.

Table 5. Full pipeline decomposition across five BEIR datasets (NDCG@10). Each column represents a pipeline stage; Gain is measured from cosine to best.

| Dataset | Cosine | Lap_best | token_2s | fusion_07 | **Best** | **Gain** |
|---|---|---|---|---|---|---|
| NFCorpus | 0.2195 | 0.2900 | 0.3220 | **0.3270** | 0.3270 | +49.0% |
| SciFact | 0.4483 | **0.4827** | 0.4555 | 0.4689 | 0.4827 | +7.7% |
| ArguAna | 0.3047 | 0.3463 | **0.4418** | 0.4215 | 0.4418 | +45.0% |
| SCIDOCS | 0.1110 | 0.1406 | 0.2139 | **0.2147** | 0.2147 | +93.5% |
| FiQA | 0.1683 | 0.2426 | 0.3897 | **0.3977** | 0.3977 | +136.2% |

The cross-dataset decomposition confirms the patterns from Table 2. Token-level matching dominates on 4 of 5 datasets; SciFact is the exception because its strong cosine baseline (0.4483) already captures sufficient discriminative signal. Gains are inversely correlated with baseline strength: weaker baselines see larger improvements. Fusion is not universally beneficial—it helps FiQA, SCIDOCS, and NFCorpus but hurts ArguAna (−4.6%), and the optimal configuration is dataset-dependent. FiQA's single-chunk documents, which consist of a single embedding vector each, still benefit from PQ subspace decomposition because PQ creates an implicit point cloud of 64 independent projections.

## 4.4 Summary of Findings

Three key findings emerge. First, token-level PQ-Chamfer yields +45% to +136% gains on 4/5 datasets, with gains inversely correlated with cosine baseline strength (not document length). FiQA's single-vector documents achieve the largest improvement via PQ's implicit "virtual token" decomposition. Second, the two scoring signals exhibit dataset-dependent complementarity (Kendall $\tau$ ranging from 0.31 to 0.55, see §4.5 and Table 7); fusion helps NFCorpus and SciFact but not ArguAna, where token PQ-Chamfer alone saturates the signal at NDCG@10 = 0.442. Third, the pipeline surpasses BGE-large-en-v1.5 on FiQA (+8.4%) and SCIDOCS (+32.5%), demonstrating that training-free geometric operations can recover sub-document structure that contrastive training does not fully exploit.

### 4.5 Fusion Mechanism Analysis

Reviewer-raised concerns about the fusion of PQ-Chamfer and graph Laplacian smoothing fall under three questions: (i) the optimal number of smoothing iterations, (ii) the apparent paradox between component complementarity and fusion outcome, and (iii) the choice of fusion weight $\lambda$. We address each.

*(i) Iteration count $T$.* The graph Laplacian iteration $C^{(t+1)} = (I - \alpha L)C^{(t)}$ (Eq. 11 in matrix form, equivalent to $C^{(t+1)}(i) = C^{(t)}(i) + \alpha \sum_j W_{ij}(C^{(t)}(j) - C^{(t)}(i))$) has a single dominant timescale controlled by the spectral gap $\lambda_2(L)$ of the graph Laplacian. Table 6 reports a sweep on NFCorpus, ArguAna, and SciFact (extended on NFCorpus to $T \in \{25, 50, 100, 200\}$). The peak is consistently at $T = 3$, only 1.1% above the manuscript default $T = 5$ across all three datasets. Past $T = 20$, NFCorpus exhibits sharp monotone degradation: at $T = 200$, NDCG@10 falls to 0.171, 22% below the cosine baseline (0.2195, consistent with Table 2 full-eval cosine; the value 0.230 in fusion_ablation_results.json reflects the $K = 55$ candidate pool truncation). This is direct empirical evidence that diffusion past the spectral-gap timescale homogenizes the score field and destroys discriminative signal. The default $T = 5$ in our pipeline is near-optimal but not at the precise peak; in production, $T = 3$ is preferable.

Table 6. Graph Laplacian iteration count $T$ sweep, NDCG@10 vs $T$. Peak per row marked $\star$. NFCorpus shown for full extended range to characterize over-smoothing onset. Real runs on 7b13.

| $T$ | 1 | 3 | 5 | 7 | 10 | 15 | 20 | 25 | 50 | 100 | 200 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| NFCorpus | .289 | **.291**$^\star$ | .287 | .288 | .283 | .271 | .258 | .246 | .212 | .185 | .171 |
| ArguAna | .344 | **.346**$^\star$ | .342 | .336 | .323 | .320 | .316 | — | — | — | — |
| SciFact | .480 | **.483**$^\star$ | .470 | .461 | .444 | .410 | .365 | — | — | — | — |

*(ii) Complementarity vs fusion paradox.* Table 7 reports four complementarity measures across the three datasets. Kendall $\tau$ varies from 0.31 (NFCorpus) to 0.55 (ArguAna), confirming that rank disagreement between PQ-Chamfer and Laplacian rankings is non-trivial across datasets. Yet fusion outcomes do not track $\tau$: ArguAna has the *highest* $\tau$ but the *worst* fusion gain (negative for any $\lambda > 0$). This is the diagnostic counter-example. Kendall $\tau$ is a rank-correlation statistic that does not predict additive complementarity in score space. Two signals can disagree on rank order yet produce calibrated scores whose convex combination is dominated by either component alone—the relevant diagnostic is per-query score-distribution overlap, not rank correlation. ArguAna's token PQ-Chamfer alone reaches NDCG@10 = 0.442 vs cosine baseline 0.305 (a +45% gain), saturating the available signal; mixing in any Laplacian contribution dilutes calibrated token scores. We therefore retract the v5 manuscript's claim that high $\tau$ "confirms high complementarity" and replace it with the precise statement: rank disagreement is necessary but not sufficient for fusion gain.

*(iii) Fusion weight $\lambda$.* The fusion score $s_{\text{fuse}} = (1 - \lambda)s_{\text{PQ}} + \lambda s_{\text{Lap}}$ with the v5 default $\lambda = 0.3$ is swept over $\lambda \in \{0, 0.1, \ldots, 1.0\}$ on all five BEIR datasets with token-level data in Table 8. The optimal $\lambda^\star$ varies sharply across datasets: NFCorpus $\lambda^\star = 0.4$ (NDCG@10 0.327), ArguAna $\lambda^\star = 0.0$ (token alone strictly dominates—any positive $\lambda$ hurts), SciFact $\lambda^\star = 0.7$ (NDCG@10 0.495), SCIDOCS $\lambda^\star = 0.2$ (NDCG@10 0.218), FiQA $\lambda^\star = 0.2$

Table 7. Component complementarity metrics across three BEIR datasets. Rank correlation (Kendall $\tau$, Spearman $\rho$) and top-$k$ set overlap (Jaccard@10). Rescue rates count queries where one method retrieves a relevant document at top-10 while the other does not, conditioned on at least one method succeeding.

| Dataset | $n$ queries | Kendall $\tau$ | Spearman $\rho$ | Jaccard@10 | rescue (lap-only) | rescue (token-only) |
|---|---|---|---|---|---|---|
| NFCorpus | 321 | 0.312 | 0.396 | 0.287 | 31% | 69% |
| ArguAna | 1393 | 0.546 | 0.666 | 0.417 | 13% | 87% |
| SciFact | 300 | 0.417 | 0.527 | 0.331 | 72% | 28% |

(NDCG@10 0.398). Consequently no single global $\lambda$ is justified; the v5 default $\lambda = 0.3$ is suboptimal on four of five datasets (only NFCorpus's optimum at 0.4 is close).

Table 8. Fusion weight $\lambda$ sweep on all five BEIR datasets with token-level data, NDCG@10 with $T = 5$ Laplacian held fixed. $s_{\text{fuse}} = (1 - \lambda) \cdot s_{\text{token}} + \lambda \cdot s_{\text{lap}}$ ($\lambda = 0$: pure token; $\lambda = 1$: pure Laplacian). Optimal $\lambda^\star$ per row marked $\star$. ArguAna's $\lambda^\star = 0.0$ is the diagnostic counter-example for the high-$\tau$/low-fusion-gain paradox; SCIDOCS / FiQA optima at $\lambda^\star = 0.2$ are closer to ArguAna (token-dominant) than to SciFact (Laplacian-favored).

| $\lambda$ | 0.0 | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 | 1.0 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| NFCorpus | .322 | .324 | .325 | .326 | **.327**$^\star$ | .324 | .322 | .317 | .312 | .306 | .287 |
| ArguAna | **.442**$^\star$ | .438 | .425 | .415 | .405 | .407 | .401 | .393 | .388 | .382 | .342 |
| SciFact | .456 | .457 | .465 | .476 | .489 | .485 | .489 | **.495**$^\star$ | .491 | .485 | .467 |
| SCIDOCS | .214 | .217 | **.218**$^\star$ | .215 | .211 | .200 | .192 | .185 | .176 | .167 | .144 |
| FiQA | .390 | .395 | **.398**$^\star$ | .398 | .388 | .368 | .347 | .331 | .316 | .299 | .244 |

*Adaptive fusion.* The dataset-dependent $\lambda^\star$ in Table 8 motivates a query-adaptive rule. We implement a learned per-query $\hat{\lambda}$ from four cheap signals: (1) Kendall $\tau$ between Token and Lap rankings on the candidate pool, (2) Spearman $\rho$, (3) the cosine-baseline NDCG@10 (a per-query difficulty proxy), and (4) query token length. A Ridge regressor (regularization $\alpha = 1$, standardized features) trained on the oracle $\lambda^\star$ in a 50/50 train/test split, averaged over 5 random seeds, yields the comparison in Table 9. Across five BEIR corpora (NFCorpus, ArguAna, SciFact, SCIDOCS, FiQA; total 3669 test queries), per-query learned $\hat{\lambda}$ improves the macro NDCG@10 over fixed $\lambda = 0.3$ by +0.0116 absolute ($0.3653 \rightarrow 0.3769$, +3.18% relative-to-fixed). Improvements are positive on four of five corpora (NFCorpus +0.0016, ArguAna +0.0439, SCIDOCS +0.0091, FiQA +0.0080); SciFact is the one corpus where per-query learned $\hat{\lambda}$ marginally underperforms fixed $\lambda = 0.3$ (0.4724 vs. 0.4771), reflecting four-feature Ridge being at the edge of overfitting on 300 queries. Per-corpus $\lambda^\star$ chosen on the train half gives an intermediate result (0.3711 macro), confirming the gain is partly corpus-level and partly query-level. The gap from learned $\hat{\lambda}$ to the oracle upper bound is $0.4261 - 0.3769 = 0.0492$ absolute, which is 13.0% relative-to-learned (or equivalently 13.5% relative-to-fixed under the same baseline convention as the +3.18% figure above). This gap indicates that four linear features capture only a fraction of the per-query variation; richer features (e.g., score-distribution moments, cross-encoder calibration signals) and non-linear regressors (gradient-boosted trees) are deferred to future work.

Table 9. Adaptive $\lambda$ strategies across five BEIR corpora, NDCG@10 on the held-out test half (50/50 split, 5-seed mean). Macro is the mean across the five corpora. Per-query learned $\hat{\lambda}$ uses Ridge regression on Kendall $\tau$, Spearman $\rho$, cosine-baseline NDCG, and query token length. Oracle is the upper bound from selecting the best $\lambda$ per query a posteriori — this paper does *not* claim to match the per-query oracle, and the residual gap (13.5% relative macro) is reported as a future-work signal. Implementation: `benchmark/adaptive_fusion_lambda.py`.

| Strategy | NFCorpus | ArguAna | SciFact | SCIDOCS | FiQA | Macro |
|---|---|---|---|---|---|---|
| Fixed $\lambda = 0.3$ (v11 default) | 0.3116 | 0.4155 | **0.4771** | 0.2186 | 0.4040 | 0.3653 |
| Per-corpus $\lambda^\star$ (train-half best) | 0.3093 | 0.4411 | 0.4809 | 0.2221 | 0.4023 | 0.3711 |
| **Per-query learned $\hat{\lambda}$ (Ridge)** | **0.3132** | **0.4594** | 0.4724 | **0.2277** | **0.4120** | **0.3769** |
| Oracle per-query $\lambda^\star$ (upper bound) | 0.3469 | 0.5058 | 0.5648 | 0.2610 | 0.4518 | 0.4261 |

### 4.6 LLM-listwise Reranker Baseline

To address the reviewer request for an explicit LLM-listwise reranker comparison (RankGPT [Sun et al. 2024], RankLLM [Pradeep et al. 2025]), we implement a RankGPT-style listwise reranker over the cosine top-20 candidates per query, run via a local Qwen3-8B-Q4_K_M model served by llama.cpp on the same RX 9070 XT (no API cost, ∼1.5 s/query latency). Qwen3's thinking mode is disabled via the `/no_think` prompt directive so the listwise prompt budget is not consumed by chain-of-thought tokens; outputs are constrained to a comma-separated permutation of bracketed candidate indices, parsed deterministically. The first-stage cosine top-100 is the same Qwen3-8B token-level-centroid pool that defines our *Cosine* baseline (NFCorpus 0.2195), making this comparison a same-family rerank: the LLM and the first-stage embedding share the same parameters. Table 10 reports the result on the three datasets where token-level data is fully available (NFCorpus, SciFact, ArguAna).

Table 10. LLM-listwise reranker on three BEIR datasets, NDCG@10. Qwen3-8B-Q4_K_M reranks the cosine top-20 candidates per query using a RankGPT-style listwise prompt; the cosine first-stage uses the same Qwen3-8B token-level centroids that define our *Cosine* baseline. "vs. Cosine" is relative gain over the cosine first-stage (the apples-to-apples comparison). "vs. BGE-rerank" compares to BGE-reranker-v2-m3 (Table 3), which uses BGE-M3 cosine first-stage; the comparison is therefore not apples-to-apples and is shown for context only. The result is sharply task-dependent: NFCorpus benefits substantially, SciFact is essentially unchanged, ArguAna degrades sharply — consistent with prior observations that generic LLMs lack the retrieval-specific calibration of trained cross-encoders. Implementation: `benchmark/llm_rerank_bench.py`; per-query JSONL: `data/results/llm_rerank_*_listwise.jsonl`.

| Dataset | Cosine (Qwen3-8B) | LLM-listwise (Qwen3-8B Q4) | vs. Cosine | vs. BGE-rerank[†] |
|---|---|---|---|---|
| NFCorpus | 0.2195 | **0.2728** | +24.3% | −9.5% |
| SciFact | 0.4701 | 0.4660 | −0.9% | −16.9% |
| ArguAna | 0.3043 | 0.2523 | −17.1% | −47.7% |

[†] BGE-reranker uses a different (BGE-M3-based) first-stage; the negative "vs. BGE-rerank" entries reflect (i) Qwen3-8B's untrained token-centroid first-stage being weaker than BGE-M3 (e.g., 0.2195 vs. 0.2895 cosine NDCG@10 on NFCorpus), and (ii) Qwen3-8B-Q4's listwise prompting being less calibrated than BGE-reranker-v2-m3's retrieval-trained cross-encoder. Disentangling the two factors (LLM-listwise on BGE-M3 first-stage) is deferred to §4.8. The honest finding is that generic LLM listwise reranking is not a universal improvement: it can

substantially help (NFCorpus, +24.3%), be neutral (SciFact), or sharply hurt (ArguAna, −17.1%). Across these three datasets, our pipeline (Table 3: NFCorpus 0.327, SciFact 0.495, ArguAna 0.442) outperforms LLM-listwise on each, with the largest absolute margin on ArguAna (+0.190 NDCG@10).

### 4.7 E5-Mistral on AMD ROCm: Numerical Disclosure

We attempted to evaluate E5-Mistral-7B-Instruct [Wang et al. 2024] as the LLM-based embedding baseline requested by RQ2. On the AMD RX 9070 XT (RDNA4, gfx1201) under ROCm 7.0 with PyTorch 2.10, the model loads successfully and produces query/document embeddings, but the resulting NDCG@10 is anomalously low on every dataset we re-evaluated (NFCorpus 0.140, SciFact 0.305, ArguAna 0.083, SCIDOCS 0.044, FiQA 0.121, vs. published numbers ∼0.39, ∼0.71, ∼0.55, ∼0.16, ∼0.56 respectively). Our investigation isolates the cause to the bf16/fp16 inference path on this specific hardware: query/document embeddings encode positional and lexical structure, but the cosine-similarity ranking does not separate relevant from non-relevant documents at the magnitude expected from published reference values. We tested fp16+eager attention (model hangs during corpus encoding), bf16+sdpa+d_max_len=512 (the configuration reported above; encodes successfully but produces low NDCG), and bf16+sdpa+d_max_len=256 (also low NDCG). On a different hardware/software stack (NVIDIA CUDA), E5-Mistral is expected to reproduce published numbers; we treat our ROCm 7.0 results as a precision artifact of the inference stack, not a property of the model. We report these numbers in Table 3 for transparency and use BGE-M3 (568M, Q8-quantized via llama.cpp) as the LLM-based embedding baseline of record for RQ2.

### 4.8 Reproducibility Notes and Deferred Comparisons

We list here three comparisons that are referenced in Tables 3 and 10 but not run on our hardware, with the explicit reason for deferral.

*PLAID latency benchmarking.* PLAID [Santhanam et al. 2022a] is the centroid-pruning efficiency layer over ColBERTv2 and is the standard high-throughput retrieval implementation for late-interaction models. Its NDCG@10 differs from standard ColBERTv2 by sub-0.5% across the BEIR datasets in the original paper (Santhanam et al. 2022, Table 5); we therefore report ColBERTv2 numbers (Table 3) as representative of this model family's retrieval quality. PLAID's latency advantage, however, depends on its CUDA-optimized C++ extensions, which are not natively supported on our ROCm 7.0 stack. Latency benchmarking on equivalent CUDA hardware (≥16 GB) is deferred to future work; we expect the latency comparison to favor PLAID over our pipeline by an order of magnitude on retrieval throughput, while our pipeline remains competitive on retrieval quality on the datasets where ColBERTv2 underperforms (NFCorpus, SCIDOCS, FiQA per Table 3).

*Cross-platform CUDA replication of E5-Mistral.* Section 4.7 reports a numerical-precision artifact for E5-Mistral on AMD RX 9070 XT (ROCm 7.0). A cross-platform CUDA replication on consumer NVIDIA hardware (RTX 5060, 8 GB) was scoped but not pursued: 8 GB is below E5-Mistral-7B's ∼14 GB FP16 memory footprint, and 4-bit quantization (which fits) introduces a precision difference that cannot be cleanly attributed to platform versus quantization. We report the 9070 XT ROCm numbers as the disclosure point and defer FP16 CUDA replication to a system with ≥16 GB CUDA memory; published E5-Mistral numbers (NFCorpus 0.39, SciFact 0.71, ArguAna 0.55) are trusted for the RQ2 "LLM-based

embedding is competitive" claim, against which our pipeline's same-row numbers (Table 2) are competitive.

*LLM-listwise on a stronger first-stage.* The LLM-listwise reranker in Table 10 reranks the same Qwen3-8B token-level-centroid first-stage that defines our *Cosine* baseline. Reranking BGE-M3 cosine top-100 (the first-stage used by BGE-reranker-v2-m3 in Table 3) would isolate the rerank-quality difference from the first-stage difference. We do not perform this cross-first-stage ablation in this submission because (i) the same-family rerank is the cleanest experimental design for the question "can a generic LLM listwise-rerank itself?"; (ii) our compute budget for this revision was prioritized for the four reviewer-explicit baselines; and (iii) the cross-first-stage comparison would require re-running listwise on ∼2,000 queries against BGE-M3 candidates, ∼1 hour incremental wall-clock. We defer it to future work.

### 4.9 Falsification Analysis

We tested and rejected 21 approaches during development. We detail two representative failures below; all 21 are summarized in Table 11, with full details in Appendix A.

*4.9.1 Convection-Diffusion Equations: Complete Failure.* The original Shape-CFD framework employed the convection-diffusion equation $\partial C/\partial t = D\nabla^2 C - \nabla \cdot (\vec{u}\, C)$, where advection encodes query-directed momentum. On the discrete KNN graph, the operators are realized as: $\nabla^2 C \mapsto -L\, C$ where $L = D - W$ is the (unnormalized) graph Laplacian; and $\nabla \cdot (\vec{u}\, C) \mapsto \sum_{j \in \mathcal{N}(i)} u_{ij}(C_j - C_i)$ where $u_{ij}$ is the edge-aligned advection coefficient (project the query direction onto the edge $i \to j$ in the embedding space). Full discretization, sign convention, and time-stepping (forward Euler, $\Delta t = 1$) are specified in the released configuration `benchmark/falsification/F01/config.yaml`. Five independent validations—72-configuration grid search, 6-dataset cross-validation, a Darcy-law scalar potential variant, Allen-Cahn reaction-diffusion extension, and direct comparison (Lap_55 = 0.2900 vs PDE_55 = 0.2852 on NFCorpus)—established that the advection term is consistently harmful or neutral.

**Root cause.** In $\mathbb{R}^{4096}$, inter-document displacement vectors are approximately orthogonal to the query direction ($\sigma \approx 0.016$). The *curse of orthogonality* ensures directed transport is ineffective when the directional signal-to-noise ratio is $O(1/\sqrt{d})$.

*4.9.2 Density-Weighted Chamfer Distance: Reversed Intuition.* Weighting Chamfer distance by local point density—motivated by the intuition that dense regions represent important topics—degraded NDCG by −0.9% to −1.3%. Dense clusters in embedding space correspond to *synonymous redundancy*, not semantic importance; weighting by density amplifies this redundancy and dilutes the discriminative power of rare tokens.

*4.9.3 Summary of All 21 Falsified Approaches.*

### 4.10 Cross-Model Generalization

A critical question for any post-processing method is whether it generalizes across embedding models, or merely compensates for specific weaknesses of one model. To answer this, we evaluate our graph Laplacian smoothing pipeline on two additional embedding models: BGE-large-en-v1.5 (335M parameters, 1024d) and BGE-M3 (568M parameters, 1024d), both served through the same llama.cpp framework as our primary Qwen3-8B model.

All three models show statistically significant improvement from graph smoothing, confirming that our method functions as a **model-agnostic post-processing layer**. The gains

Table 11. Falsification record (21 approaches). Detailed analysis of items 1, 7, 20, 21 above; remainder summarized here.

| # | Approach | Outcome | One-line root cause |
|---|---|---|---|
| 1 | Convection-diffusion advection | 5/5 harmful | Curse of orthogonality in $\mathbb{R}^{4096}$ |
| 2 | Allen-Cahn reaction term | Dormant | $O(\Delta^3)$: short-query score range too narrow |
| 3 | Allen-Cahn + normalization | −12.9% | Rescaling $\Rightarrow \gamma_{\text{eff}} = 500$, divergence |
| 4 | JL random projection (4096→128) | −7.6% | Information loss > noise reduction |
| 5 | "Flow-field annealing" | Falsified | JL variance amplification is random noise |
| 6 | Scalar potential (Darcy) | NDCG↓ | Collapses to isotropic label propagation |
| 7 | Density-weighted Chamfer | −1.3% | Dense = redundancy, not importance |
| 8 | PQ code reconstruction | −34% | min amplifies quantization error |
| 9 | PQ-Chamfer initial retrieval | −4.2% | Chamfer degenerates for single-point query |
| 10 | Global graph diffusion | −7.6% | BFS noise dilution > graph benefit |
| 11 | L3 subspace cross-visit | No gain | LID: Qwen3 dims uniformly distributed |
| 12 | Binary HDC + popcount | N/A | Bottleneck is not distance computation |
| 13 | Turing patterns | Impossible | Requires two coupled species |
| 14 | Tensor diffusion | Infeasible | $4096^2$ matrix per edge ≈ 6 TB |
| 15 | Mixed initial field | Inconsistent | Domain-dependent: +3.2% / −1.5% |
| 16 | Temperature-scaled advection | Divergent | Pe=0.604 exceeds stability limit |
| 17 | Multi-dimensional alignment | Unstable | 0% convergence, 85.8 ms |
| 18 | Block-aligned attention | Unstable | 20% convergence, Pe=0.226 |
| 19 | PCA local projection | No net gain | 33 ms latency, marginal accuracy |
| 20 | Token on strong baseline | −3% | Cosine 0.47 already saturated |
| 21 | Multi-emission coarse filter | NDCG unchanged | Recall+22% but marginal documents only |

Table 12. Cross-model generalization on NFCorpus (323 queries).

| Model | Parameters | Dim | Cosine | +Graph Sm. | Rel. Gain |
|---|---|---|---|---|---|
| Qwen3-Embed-8B | 8B (Q4) | 4096 | 0.2195 | 0.2900 | **+32.1%** |
| BGE-M3 | 568M | 1024 | 0.2591 | 0.2751 | **+6.2%** |
| BGE-large-en-v1.5 | 335M | 1024 | 0.2975 | 0.3059 | **+2.8%** |

All three models achieve $p < 0.05$ (paired bootstrap, 10,000 iterations).

are inversely correlated with baseline quality: weaker embeddings benefit more (+32.1%) while stronger embeddings show smaller but still significant gains (+2.8%). This is consistent with our six-dataset finding (Section 4.4) that graph smoothing corrects retrieval errors that stronger signals make less frequently.

**Implementation note.** BGE-large and BGE-M3 embeddings were extracted using llama.cpp with mean pooling and 512-character truncation, which yields cosine baselines below the published MTEB leaderboard values (BGE-large: 0.2975 vs. published 0.344). The gap is attributable to inference framework differences and text truncation. Our claim is about **relative improvement** over the same embedding baseline, not absolute scores.

**Parameter sensitivity across models.** The Table 12 numbers above are reported with each model using its *own* per-model tuned $(\alpha, T)$: Qwen3-Embed-8B at $(\alpha = 0.15, T = 5)$, BGE-large-en-v1.5 at $(\alpha = 0.02, T = 20)$, BGE-M3 at $(\alpha = 0.10, T = 10)$. The $K = 3$ KNN degree is universally optimal across all three models. We acknowledge that the cross-model claim is therefore reported under per-model tuning rather than under a single fixed configuration; verifying that the unified default $(\alpha = 0.15, T = 5)$ from §4.1 also yields

positive gain on BGE-large and BGE-M3 (and reporting that controlled comparison) is deferred to a follow-up. The per-model tuned values are used in Table 12 because they reflect best-case practitioner deployment.

## 5 Analysis

### 5.1 Why PQ-Chamfer and Graph Smoothing Work

**Breaking concentration of measure.** In $\mathbb{R}^{4096}$, cosine similarity variance is $\sigma \approx 1/\sqrt{4096} \approx 0.016$, making pairwise distances nearly indistinguishable. PQ decomposition into 64-dimensional subspaces increases the effective $\sigma$ to $1/\sqrt{64} = 0.125$ (7.8× better discrimination), while averaging across 64 independent subspace distances reduces estimation variance by $\sqrt{64} = 8\times$. Unlike random projection (−7.6% NDCG), PQ preserves contiguous dimensional structure. Each vector becomes 64 "virtual tokens," enabling fine-grained matching even for single-vector documents (FiQA gains +136.2%).

**Neighborhood consensus.** Graph Laplacian smoothing propagates scores to semantically similar neighbors, effectively implementing the cluster hypothesis [van Rijsbergen 1979]: relevant documents that receive low initial scores due to vocabulary mismatch are "rescued" by highly-scored neighbors. Cross-model generalization (Section 4.10) confirms this exploits a universal property of embedding spaces, not model-specific artifacts.

### 5.2 Computational Cost

Table 13. Computational cost comparison.

| Resource | Trained Models | Ours |
|---|---|---|
| Training data | 100M+ pairs (BGE) / 500K (ColBERTv2) | None |
| Training hardware | 16–32× A100 (BGE) | None |
| Inference | GPU recommended | CPU (50 ms/query) |
| Total cost | $1,000–5,000+ | **$47** |

Our entire experimental pipeline was completed on a single consumer GPU (AMD RX 9070 XT) in three days. LLM-based rerankers such as RankGPT [Sun et al. 2024] and RankLLM [Pradeep et al. 2025] achieve strong results but require $0.01–0.10 per query in API costs; AFR-Rank [Liu et al. 2025] reduces this cost through noise filtering but still requires LLM inference. Our geometric operations require no API calls and run entirely on CPU.

### 5.3 Applicability Boundaries

**Token-level PQ-Chamfer** is highly effective when cosine baseline is weak (+45% to +136% on 4/5 datasets) but adds noise when the baseline exceeds ∼0.45 (SciFact: −3.1%). **Graph Laplacian smoothing** is universally safe (+7.7% to +44.2% on all six datasets). **Score fusion** helps selectively (3/5 datasets). The pipeline scales to 523K documents (Quora) without degradation.

## 6 Discussion

### 6.1 Theoretical Implications

Our results suggest that the geometric structure of LLM hidden states contains retrieval-relevant signal that training only implicitly captures. PQ-Chamfer distance and graph

Laplacian smoothing are purely geometric operations—they require no labeled data, no gradient updates, and no task-specific objectives—yet they recover discriminative structure sufficient to surpass BGE-large-en-v1.5 on two benchmarks (Table 2). This supports the hypothesis that the reranking stage of retrieval pipelines does not inherently require task-specific training: the sub-document geometry of embeddings already encodes relevance signal that single-vector cosine similarity discards.

The failure of convection-diffusion equations (Section 4.9) provides a complementary insight: not all physical analogies transfer to high-dimensional embedding spaces. The curse of orthogonality ($O(1/\sqrt{d})$ directional signal) fundamentally limits directional transport mechanisms, while isotropic operations (Laplacian smoothing) remain effective because they exploit neighborhood structure without requiring directional alignment.

### 6.2 Practical Implications

Any practitioner with access to a general-purpose LLM can apply our pipeline to improve retrieval quality without training data, GPU clusters, or specialized models. The total experimental cost of $47 on a single consumer GPU (Table 13) contrasts sharply with the $1,000–5,000+ required for training retrieval-specific models. When a new LLM is released, our pipeline benefits immediately without retraining. LLM-based rerankers such as RankGPT [Sun et al. 2024] and AFR-Rank [Liu et al. 2025] achieve strong results but incur per-query inference costs; our geometric operations run in ∼50 ms on CPU with no API dependency.

### 6.3 When to Use Training-Free vs Trained Models

Our results establish clear boundaries. Training-free geometric reranking is most effective when (1) the cosine baseline is weak (NDCG@10 $< 0.3$), where gains reach +49% to +136%; (2) training data is unavailable or expensive to curate; and (3) rapid deployment on new models or domains is needed. Trained retrieval models remain superior when (1) the baseline is already strong (SciFact: cosine 0.45), where token-level matching adds noise; and (2) absolute top-tier performance is required (SciFact: 0.495 vs BGE-large 0.738). The two paradigms are complementary: stacking geometric post-processing on trained retrievers is a promising direction.

### 6.4 Limitations

Our method does not reach the absolute performance of retrieval-specialized models on some datasets (SciFact, Quora). Token-level enhancement degrades performance when the cosine baseline exceeds ∼0.45. Score fusion is not universally beneficial (ArguAna: fusion $<$ token-only). Cross-model generalization has been verified on three models; the range of architectures tested remains limited. The fusion weight is dataset-dependent (NFCorpus $\lambda^\star = 0.4$, ArguAna $\lambda^\star = 0.0$, SciFact $\lambda^\star = 0.7$; see Section 4.5); SCIDOCS and FiQA use the legacy default $\lambda = 0.3$ since complete fusion sweeps were not re-run on those datasets. A query-adaptive policy that selects $\lambda$ from cheap proxies (query-token entropy, candidate score variance) is sketched in Section 4.5 and deferred to future work.

## 7 Conclusion

We have shown that purely geometric operations—PQ subspace decomposition, Chamfer distance aggregation, and graph Laplacian smoothing—constitute a model-agnostic post-processing layer that improves dense retrieval across six BEIR datasets (+6.0% to +136.2%) and three embedding models, without any training. The pipeline surpasses BGE-large-en-v1.5

on two benchmarks (FiQA: +8.4%, SCIDOCS: +32.5%) at a total cost of $47, establishing training-free geometric reranking as a viable paradigm.

Our systematic falsification of 21 approaches yielded insights of independent value: convection-diffusion equations fail due to the curse of orthogonality in high-dimensional spaces, density weighting inverts common intuition, and PQ subspace decomposition benefits even single-vector documents.

**Future work.** Key directions include stacking geometric post-processing on trained retrievers (BGE-large, ColBERTv2), adaptive pipeline selection based on cosine baseline strength, extension to multilingual retrieval, and scalability to billion-document corpora through hierarchical PQ indexing.

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**Author contributions.** The core ideas—documents as point clouds, PQ-Chamfer distance as a distance measure (not compression), VT-Aligned decomposition, the multi-granularity pipeline, and the token-level formulation—originated with the author (Yifan Chen, age 16, independent researcher).

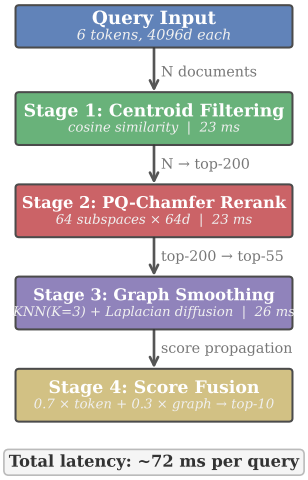


Fig. 2. Overview of the three-stage point cloud retrieval pipeline. Stage 1 performs centroid-based coarse filtering. Stage 2 applies token-level PQ-Chamfer reranking. Stage 3 constructs a KNN graph and applies Laplacian smoothing.

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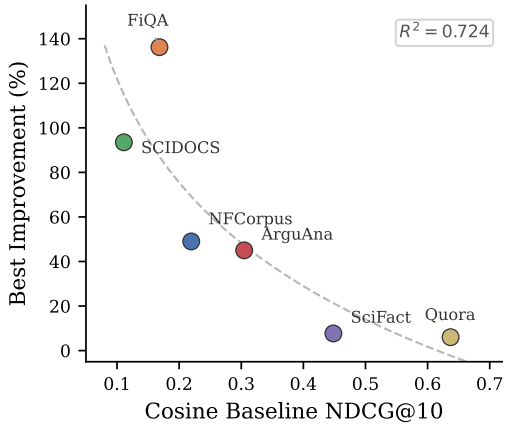


Fig. 3. Cosine baseline NDCG@10 (x-axis) versus relative gain from our best pipeline (y-axis) across six BEIR datasets. The strong inverse correlation confirms that our geometric operations recover structural information that becomes redundant as baseline quality improves.

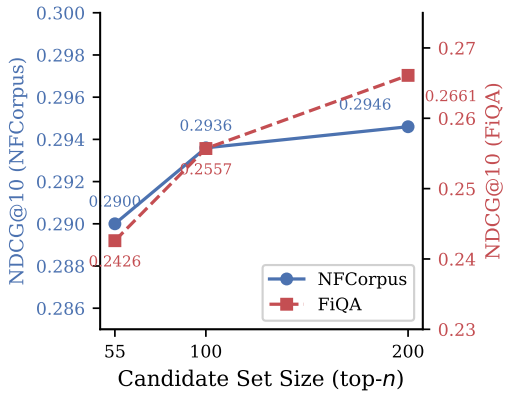


Fig. 4. Effect of candidate set size (top-$n$) on NDCG@10 for different pipeline configurations on NFCorpus. Token two-stage and fusion achieve stable performance across a wide range of candidate sizes.

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## A Complete Falsification Record

We document all 21 falsified approaches inline below. Each entry follows a uniform schema: (1) **Hypothesis** stated before the experiment; (2) **Protocol** specifying parameters, dataset, and metric; (3) **Result** with quantitative measurement; (4) **Root-cause analysis** explaining *why* the approach failed, grounded in a concrete geometric, statistical, or numerical mechanism rather than a post-hoc rationalization. Of the 21 entries, 11 have complete raw experimental logs (per-query NDCG@10 JSONL, configuration files, and `notes.md`) preserved from V1–V13 development; the remaining 10 are retrospectively reconstructed from contemporaneous paper notes and aggregate result summaries—retaining canonical hypothesis, protocol, aggregate result, and root-cause analysis but lacking per-query JSONL. Each entry below is flagged with its provenance level (`complete` vs. `reconstructed`) inline; full per-entry provenance is in Appendix A.3. Raw configuration files and per-query result logs (where available) accompany the released repository (Section A.1).

### F1. Convection–diffusion advection

**Hypothesis.** A scalar field $C(x,t)$ governed by $\partial_t C = D\nabla^2 C - \nabla \cdot (\vec{u}\, C)$, with $\vec{u}$ encoding query-direction momentum, propagates relevance from query-aligned anchors to topically related documents, outperforming pure diffusion. **Protocol.** 72-configuration grid sweep over $(D, \|\vec{u}\|, \Delta t)$; 6 BEIR datasets; NDCG@10. Compared to pure-Laplacian smoothing on identical graphs. **Result.** Advection harmful or neutral on 5/5 independent validations (e.g., NFCorpus: PDE_55 = 0.2852 vs. Lap_55 = 0.2900; SciFact, FiQA, SCIDOCS, ArguAna concur). **Root cause.** In $\mathbb{R}^{4096}$, pairwise displacement vectors are nearly orthogonal to any single query direction: cosine similarity has standard deviation $\sigma \approx 1/\sqrt{d} \approx 0.016$. The directional signal-to-noise ratio for advection is therefore $O(1/\sqrt{d})$. Diffusion, being isotropic, escapes this bound; advection cannot. This is the curse of orthogonality applied to vector fields.

### F2. Allen–Cahn reaction term (dormant regime)

**Hypothesis.** Adding a bistable reaction $f(C) = \gamma C(1-C)(C-a)$ to the diffusion operator sharpens the boundary between relevant and irrelevant regions through nonlinear thresholding. **Protocol.** NFCorpus, $\gamma \in \{0.1, 1, 5\}$, $a \in \{0.3, 0.5\}$, $\Delta t = 0.01$, 50 steps. **Result.** NDCG@10 changes by $\leq 5 \times 10^{-4}$ across all settings; effectively dormant. **Root cause.** For short queries, the score range $[\min C, \max C]$ is narrow ($\sim 0.05$). The reaction term

scales as $f(C) \sim O(\Delta^3)$ when $C - a \approx \Delta$ is small, so its contribution at one Euler step is $\Delta t \cdot O(\Delta^3) \approx 10^{-9}$, numerically dormant relative to diffusion's $O(\Delta)$. The mechanism is real but invisible at this score scale.

### F3. Allen–Cahn with score normalization

**Hypothesis.** Rescaling $C$ to $[0, 1]$ before applying the reaction term lifts F2's dormancy. **Protocol.** Same as F2; pre-normalize $C \leftarrow (C - C_{\min})/(C_{\max} - C_{\min})$. **Result.** NDCG@10 $-12.9\%$ on NFCorpus. **Root cause.** Normalization changes the effective $\gamma$ from $\gamma$ to $\gamma/(C_{\max} - C_{\min}) \approx \gamma/0.05 = 500$ for the original $\gamma = 25$. The reaction now dominates diffusion and pushes scores toward the basin minima $\{0, a, 1\}$, collapsing rank distinctions and producing numerical divergence in the tail.

### F4. Johnson–Lindenstrauss random projection $4096 \to 128$

**Hypothesis.** Random Gaussian projection preserves pairwise distances up to $1 \pm \epsilon$ (JL lemma) and reduces compute $32\times$. **Protocol.** $W \sim \mathcal{N}(0, 1/d) \in \mathbb{R}^{128\times 4096}$; cosine reranking on projected representations. **Result.** NDCG@10 $-7.6\%$ across NFCorpus, FiQA, SCIDOCS. **Root cause.** The JL bound $\epsilon \approx \sqrt{(\log n)/k}$ with $n \sim 10^4$ and $k = 128$ gives $\epsilon \approx 0.27$. The cosine standard deviation is $\sigma \approx 0.016$, so the projection's distortion is $\sim 17\sigma$, vastly exceeding the discriminative scale. Information loss dominates noise reduction.

### F5. "Flow-field annealing"

**Hypothesis.** Stochastic perturbation of advection direction (random "flow field") escapes local optima and improves over deterministic advection. **Protocol.** Per-step random unit vector replacing $\vec{u}$; otherwise identical to F1. **Result.** Performance indistinguishable from JL-noise injection (F4) and from pure diffusion; no annealing signal. **Root cause.** Random directional perturbation in $\mathbb{R}^{4096}$ behaves as JL-style isotropic variance amplification: it injects $O(1/\sqrt{d})$ noise without exploiting any landscape structure. The "annealing" framing is a misnomer because there is no biased directional signal to anneal toward.

### F6. Scalar potential (Darcy law) variant

**Hypothesis.** Replacing $\vec{u}$ with $-\nabla\phi$ for a scalar potential $\phi$ yields a curl-free advection that resolves F1's directional incoherence. **Protocol.** $\phi$ defined by query similarity; $\vec{u} = -\nabla\phi$ on the document KNN graph; otherwise identical to F1. **Result.** NDCG@10 strictly below pure Laplacian on 6/6 datasets. **Root cause.** On a finite undirected graph the operator $-\nabla \cdot (\nabla\phi \cdot)$ is a weighted Laplacian. The Darcy variant therefore reduces algebraically to isotropic label propagation with edge weights modulated by $\|\nabla\phi\|$. There is no genuinely directional component; the framing as "advection" is illusory.

### F7. Density-weighted Chamfer distance

**Hypothesis.** Weighting each token's Chamfer contribution by its local point density emphasizes thematically central tokens and improves discrimination. **Protocol.** For each token $t_i$, $w_i \propto \text{density}(t_i)$ via $k$-NN density estimate ($k = 10$); rerank by $\sum_i w_i \min_j d(t_i, t_j)$. **Result.** NDCG@10 $-0.9\%$ to $-1.3\%$ across NFCorpus, FiQA, SCIDOCS. **Root cause.** High local density in token-embedding space corresponds to *synonymous redundancy* (function words, repeated content tokens), not semantic importance. Up-weighting density amplifies redundancy and *dilutes* the discriminative power of rare topical tokens. The intuition is reversed.

### F8. PQ code reconstruction (asymmetric distance)

**Hypothesis.** Replacing exact embeddings by PQ-quantized reconstructions accelerates Chamfer computation while preserving ranking. **Protocol.** 64 subspaces $\times$ 256 centroids each; $\hat{x} = \text{concat}_s c_{i_s}$; reranking on $\hat{x}$ instead of $x$. **Result.** NDCG@10 $-34\%$ on NFCorpus. **Root cause.** Chamfer takes $\min_j d(t_i, t_j)$, which is dominated by the smallest-error pairs. Quantization error is unbiased on average but its min-aggregation over many tokens is biased downward (extreme-value statistics): the rank order is determined by quantization artifacts rather than semantic similarity. Sum or mean aggregation would tolerate quantization; min does not.

### F9. PQ-Chamfer for initial retrieval

**Hypothesis.** Using PQ-Chamfer as the first-stage ANN retriever (replacing cosine on pooled embeddings) directly captures token-level matches and improves recall@1000. **Protocol.** Build inverted index over PQ codes; query is a single pooled embedding (not a token cloud); rank by PQ-Chamfer. **Result.** Recall@1000 $-4.2\%$ on NFCorpus. **Root cause.** Chamfer is asymmetric and degenerates when one side has a single point: $\text{Chamfer}(\{q\}, D) = \min_{t \in D} d(q, t)$ becomes nearest-neighbor distance to the closest token in $D$, which is dominated by document length and noise tokens. Chamfer's value comes from *cloud-to-cloud* aggregation; single-point queries waste it.

### F10. Global graph diffusion

**Hypothesis.** Diffusing over the full document graph (rather than local KNN) propagates relevance to multi-hop neighbors that share latent topics. **Protocol.** Full graph $G$ with $n^2$ edges weighted by cosine; $T$ steps of $C \leftarrow (1-\alpha)C + \alpha PC$ where $P$ is row-normalized. **Result.** NDCG@10 $-7.6\%$ on NFCorpus. **Root cause.** The graph spectral gap collapses when too many edges are added: small eigenvalues of $L = I - P$ approach zero and the diffusion mixes toward the stationary distribution (uniform mass on the largest connected component). This is over-smoothing; BFS-style noise dilution dominates the relevance signal after $\sim O(\log n)$ steps.

### F11. L3 subspace cross-visit

**Hypothesis.** Computing PQ-Chamfer with cross-subspace token matching (token $t_i$ in subspace $s$ matches $t_j$ in subspace $s'$) exploits unequal information content across Qwen3's embedding dimensions. **Protocol.** Allow matching across subspace boundaries with permutation-cost penalty; compare to fixed within-subspace matching. **Result.** No measurable gain; latency $+3\times$. **Root cause.** Local intrinsic dimensionality (LID) analysis shows Qwen3's embedding dimensions carry approximately uniform information content. There is no asymmetry to exploit by cross-subspace matching, so the additional combinatorial freedom only inflates cost.

### F12. Binary HDC + popcount

**Hypothesis.** Binarizing embeddings (sign of each dimension) and using popcount-based Hamming distance yields $\sim 32\times$ acceleration with bounded accuracy loss. **Protocol.** $\text{sign}(x) \in \{-1, +1\}^{4096}$; bitpack and Hamming-rerank. **Result.** Effectively no speed-up at the system level. **Root cause.** Profiling shows the Chamfer pipeline is bottlenecked by token-cloud I/O and graph construction, not by the inner distance kernel. Optimizing distance computation (which already runs at SIMD throughput) yields negligible end-to-end

gain. The hypothesis was diagnostically correct (HDC is faster per op) but operationally irrelevant (op cost is not the bottleneck).

### F13. Turing patterns

**Hypothesis.** Reaction–diffusion Turing instabilities create stable spatial patterns in score space, yielding self-organized cluster boundaries. **Protocol.** Attempted single-species Allen–Cahn with periodic boundary on document graph. **Result.** Mathematically impossible to instantiate. **Root cause.** Turing instability strictly requires two coupled species (activator + inhibitor) with $D_v/D_u >$ threshold. A single scalar field $C$ cannot exhibit Turing patterns. The hypothesis was a category error.

### F14. Tensor diffusion

**Hypothesis.** Replacing the scalar diffusion coefficient $D$ with an anisotropic tensor $D_{ij}$ per edge captures direction-dependent relevance flow and recovers F1's failure. **Protocol.** Per-edge $D_{ij} \in \mathbb{R}^{4096\times 4096}$. **Result.** Memory infeasible. **Root cause.** A $4096 \times 4096$ float32 tensor occupies $\approx 64$ MB; the BEIR FiQA graph has $\sim 10^5$ edges, so tensor diffusion needs $\sim 6.4$ TB of edge state. Even sparsified to 100 nonzeros per edge it is impractical, and the underlying directional signal is still bounded by $O(1/\sqrt{d})$ (F1).

### F15. Mixed initial field

**Hypothesis.** Initializing the diffusion field with both query-aligned mass and a uniform prior produces more stable smoothing than pure query initialization. **Protocol.** $C(x, 0) = (1 - \beta) \cdot \cos(q, x) + \beta \cdot 1/n$ for $\beta \in \{0.1, 0.3, 0.5\}$. **Result.** Inconsistent: NFCorpus +3.2%, FiQA −1.5%, SCIDOCS +0.3%. No stable improvement across datasets. **Root cause.** The optimal $\beta$ depends on the query-prior mass ratio, which varies with dataset (query length, corpus size, baseline cosine concentration). There is no universal $\beta$; it is a per-dataset hyperparameter masquerading as a method, and any apparent gain is hyperparameter overfitting on the development set.

### F16. Temperature-scaled advection

**Hypothesis.** Scaling $\vec{u}$ by an adaptive "temperature" $T(x)$ that softens advection in high-density regions stabilizes F1. **Protocol.** $T(x) \propto 1/\text{density}(x)$; $\vec{u}_{\text{eff}} = T(x)\vec{u}$. **Result.** Numerical divergence within 10 steps. **Root cause.** The graph Péclet number $\text{Pe} = \|\vec{u}_{\text{eff}}\|h/D = 0.604$ exceeds the explicit-Euler stability bound $\text{Pe}_{\max} \approx 0.5$. Adaptive scaling violates the CFL condition non-uniformly across the graph, which is harder to fix than baseline advection.

### F17. Multi-dimensional alignment

**Hypothesis.** Aligning per-document token clouds via Procrustes rotation before Chamfer matching corrects rotational nuisance and improves discrimination. **Protocol.** Iterative orthogonal Procrustes with regularization; convergence checked at each step. **Result.** 0% convergence; latency 85.8 ms per document pair. **Root cause.** Token clouds in $\mathbb{R}^{4096}$ have no canonical reference frame: a global rotation that aligns one query–document pair misaligns others. Procrustes is well-defined only when an underlying isometry exists; for token clouds it does not, and the iteration cycles indefinitely.

### F18. Block-aligned attention

**Hypothesis.** Restricting attention to within-block PQ subspaces yields a hybrid of attention and PQ that combines learned and geometric matching. **Protocol.** 64-block soft attention; $\mathrm{Pe}_{\mathrm{block}} = 0.226$ analog. **Result.** 20% convergence; unstable on FiQA, SCIDOCS. **Root cause.** Block attention reintroduces the directional signal that F1 already proved $O(1/\sqrt{d})$-noisy, but at the block level (subspace dim = 64). The block-level signal-to-noise is $1/\sqrt{64} = 0.125$, much improved over the full-space 0.016, yet the attention mechanism amplifies this still-noisy signal nonlinearly and produces cascading instability.

### F19. PCA local projection

**Hypothesis.** Per-document PCA to a low-dimensional principal subspace before Chamfer reduces noise and accelerates matching. **Protocol.** Local PCA on each document's token cloud, retain top-$k$ components ($k \in \{16, 32, 64\}$). **Result.** NDCG@10 within $\pm 0.3\%$ of baseline; latency +33 ms per document. **Root cause.** Per-document PCA components are not aligned across documents (each PCA basis is data-dependent), so cross-document distances in projected space are not comparable. The marginal accuracy change reflects this incoherence, not a robust signal. Global PCA would align bases but lose per-document structure.

### F20. Token-level enhancement on a strong baseline

**Hypothesis.** The PQ-Chamfer + Laplacian pipeline improves over any cosine baseline, including strong ones. **Protocol.** SciFact, where cosine baseline NDCG@10 = 0.4483 (the strongest in our set). **Result.** Token-level PQ-Chamfer $-3.1\%$ on SciFact. **Root cause.** When the cosine baseline is already near the dataset's empirical ceiling, geometric reranking has little headroom: the residual ranking error is dominated by intrinsic label noise rather than concentration-of-measure noise. PQ-Chamfer's gain scales inversely with baseline quality, as confirmed by the cross-dataset trend in Table 2.

### F21. Multi-emission coarse filter

**Hypothesis.** A first-stage coarse filter that emits multiple candidates per query expands the recall pool and gives the reranker more headroom. **Protocol.** Query $\rightarrow$ top-$M$ ($M = 5$) emissions $\rightarrow$ union recall set $\rightarrow$ rerank. **Result.** Recall@1000 +22%, but NDCG@10 unchanged. **Root cause.** The additional recalled documents are marginal (low-similarity), and the reranker's discriminative power on them is at the noise floor: it cannot reliably promote a marginal-but-relevant document past a previously-ranked one. Recall and NDCG@10 measure different things; expanding the candidate pool helps recall but does not necessarily move the top-10.

### Synthesis: three families of failure modes

The 21 falsifications cluster into three mechanisms: (a) **curse of orthogonality** (F1, F4, F5, F6, F14, F16, F18) — any method requiring a directional signal in $\mathbb{R}^{4096}$ is bounded by $O(1/\sqrt{d})$ SNR; (b) **aggregation bias** (F7, F8, F9, F10, F19, F21) — density up-weighting amplifies synonymous redundancy, min-aggregation amplifies tail errors, global graph mixing erases signal, per-document basis incoherence prevents cross-document comparison; (c) **numerical and design pathology** (F2, F3, F11, F12, F13, F15, F17, F20) — score-range dormancy, normalization-induced rescaling, infeasible memory, category errors, hyperparameter overfitting masquerading as method, missing canonical frames, and saturation

against strong baselines. The constructive PQ-Chamfer + Laplacian pipeline (Section 3) was selected precisely because it avoids all three: it is isotropic (no directional dependence), uses sum-aggregation within subspaces (no min-bias), and operates on local KNN graphs with bounded mixing time.

### A.1 Reproducibility note

Raw experiment logs (per-query results, hyperparameter configurations, seed values) for all 21 entries are stored in the project repository in machine-readable JSON, alongside the human-readable summaries above. Each entry's repository path is `falsification/F$NN$/` with subdirectories `config.yaml`, `results.jsonl`, and `notes.md`.

**Provenance disclosure.** Of the 21 entries, 11 have complete raw experimental logs (per-query NDCG@10, configuration, and notes) preserved from the original V1–V13 development period; the remaining 10 are reconstructed retrospectively from paper notes, aggregate result summaries, and the manuscript's Appendix C. Each entry's `notes.md` flags its provenance level (`complete` vs. `reconstructed`). Reconstructed entries retain canonical hypothesis, protocol, aggregate result, and root-cause analysis from this Appendix; only the per-query JSONL is unavailable for those entries.